



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE TITLE & CODE: DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

Title of the project	HarvestIQ – Data-Driven Crop Yield Prediction	
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Abstract

Agriculture faces challenges in accurately predicting crop yield due to variations in rainfall, temperature, soil quality, fertilizer usage, crop type, and regional conditions. Traditional methods often rely on historical data and manual estimation, which may not capture these changing factors effectively.

Existing crop yield prediction solutions use statistical and machine learning techniques to analyze agricultural and environmental factors. Models such as Linear Regression, Decision Trees, Random Forest, and Support Vector Regression have been used to estimate crop productivity from historical agricultural data.

The proposed system will use Linear Regression, Decision Tree Regressor, and Random Forest Regressor to predict crop yield. The dataset will undergo data preprocessing, exploratory data analysis, feature engineering, and model evaluation using MAE, RMSE, and R^2 score. The results will be presented through an interactive dashboard.

In the future, the system can be extended by incorporating real-time weather data, soil nutrients, irrigation information, satellite imagery, and pest or disease data. Advanced machine learning and deep learning models can also be integrated to improve prediction accuracy and provide better agricultural recommendations.